No. of Functional Features Included in the Solution

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter N, P, K, temperature, humidity, pH, rainfall	Frontend UI	Done	High
2	Input Validation	Ensures valid and correct data is entered	Backend Logic	Done	Medium
3	Crop Prediction Model	ML model predicts suitable crop based on input	ML Module	Done	High
4	Backend Processing	Processes input and connects to ML model	Flask Backend	Done	High
5	Result Display	Displays predicted crop clearly to user	Frontend UI	Done	High
6	Dataset Handling	Loads and processes crop dataset	Data Module	Done	Medium
7	Weather API Integration	Provides real-time weather data	API Integration	In Progress	High
8	UI Enhancement	Improves user interface and usability	Frontend UI	In Progress	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	Input Form, ML Prediction, Backend Processing, Result Display

Additional / Nice-to-Have Features	Weather API, UI Enhancement
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Input, Output display, UI design	Input form, result page
2	Backend / Logic	Data processing, prediction	Flask API, ML model
3	Database / Storage	Dataset storage	Flask API, ML model
4	API / Integration	External data sources	Weather API
5	Security / Authentication	Basic validation	Input validation